The Spatial Correlation Signature of Zone Formation: Interface Sharpening, Not Front Propagation

Gabriel Aubin-Moreau

2026

Abstract

Paper 21 established that the buildup timescale scales as $\tau_{\text{buildup}} \propto \nu^{-0.40}$, attributing the correction to birth-mediated spatial propagation. Here we directly measure the within-zone spatial correlation length $\xi(t)$ from the column-direction autocorrelation $C(r,t)$ of zone-mean-subtracted fieldM. The result contradicts the naive "growing front" hypothesis: $\xi(t)$ *decreases* from ∼20 sites at early times (when sg4 ≈ 0) to an equilibrium $\xi_\infty \approx 2$–3 sites as zone differentiation proceeds. This anti-correlation between sg4 and ξ identifies zone formation as *interface sharpening* (phase-separation-like), not wavefront propagation (Fisher-KPP-like): zones differentiate simultaneously across the grid, and the zone boundaries sharpen as differentiation proceeds. The equilibrium correlation length $\xi_\infty \approx 2.7$ sites is nearly independent of ν ($\xi_\infty \propto \nu^{0.04}$, $R^2 = 0.19$), confirming that ξ_∞ is set by the zone geometry (5-site zone width), not by $D_{\text{copy}}(\nu)$. The growth law exponent β in $\xi \sim t^\beta$ cannot be extracted from this observable because $\xi(t)$ is monotonically *decreasing*, not increasing.

1 Introduction

Paper 21 measured the spatial formation factor $C = \tau_{\text{buildup}}/\tau_{\text{base}}$ and found $C \propto \nu^{-0.40}$, attributing the buildup gap to birth-mediated copy-forward propagation. Two competing mechanisms predict different growth laws for the spatial correlation length $\xi(t)$ during Phase 1:

- **Diffusive propagation** ($\beta = 0.5$): $\xi \sim \sqrt{D_{\text{copy}} \cdot t}$, implying $D_{\text{copy}} \propto \nu^{0.80}$ to match $C \propto \nu^{-0.40}$.
- **Directed propagation** ($\beta = 1.0$): $\xi \sim v_{\text{copy}} \cdot t$, implying $v_{\text{copy}} \propto \nu^{0.40}$.

Measuring $\xi(t)$ would discriminate these mechanisms. We compute $C(r,t) = \langle F_{\text{res}}(x) \cdot F_{\text{res}}(x+r)\rangle / \langle |F_{\text{res}}(x)|^2\rangle$ where F_{res} is zone-mean-subtracted fieldM, and extract $\xi(t)$ via $C(r) = \exp(-r/\xi)$.

2 Methods

Protocol. Five ν values as in Paper 21: $\nu \in \{0.0005, 0.001, 0.002, 0.004, 0.008\}$, five seeds each, Phase-1-only protocol, $T_{\text{end}} = 3000$, checkpoints every 200 steps. Standard parameters: $H = 20$, HALF $= 20$, $N_{\text{zones}} = 4$, zone width 5 sites.

Autocorrelation measurement. At each checkpoint:

1. Subtract zone mean from each site: $F_{\text{res}}[i] = F[i] - \overline{F}_{\text{zone}}$.

2. Reshape to (H, HALF, H_S) grid.

3. Compute $C(r) = \langle F_{\mathrm{res}}[:, c, :] \cdot F_{\mathrm{res}}[:, c+r, :] \rangle / \mathrm{var}(F_{\mathrm{res}})$ for $r = 0 \ldots \mathrm{HALF}/2 = 10$.

Correlation length extraction. ξ is extracted via log-linear regression of $C(r)$ vs. r for $r = 1 \ldots \min(\mathrm{zone_w}, 4)$, restricted to $C(r) > 0.05$. Spurious values $\xi > 30$ (from near-zero F at early times) are excluded from trajectory plots; mean values use the last four checkpoints.

3 Results

3.1 C(r) transitions from flat to steeply decaying as zones form

Figure 1A shows the mean autocorrelation $C(r, t)$ for $\nu = 0.001$ at six checkpoints. At $t = 200$ (before zone structure has formed), $C(r) \approx 1$ for all r: the within-zone field is spatially homogeneous (F ≈ 0 everywhere after zone-mean subtraction), so every pair of sites within a zone appear perfectly correlated. This early plateau is not a sign of long-range order; it is a trivially large "correlation" of near-zero values.

As the simulation progresses and zone differentiation accelerates (sg4 rising steeply after $t \sim 1500$), $C(r)$ transitions to a sharply decaying profile: $C(1) \approx 0.7$, $C(2) \approx 0.3$, $C(3) \approx 0.1$ by $t = 3000$. This reflects the development of genuine spatial structure *within* each zone: zone boundaries create field gradients that decorrelate nearby sites within the zone.

3.2 Correlation length decreases during zone formation

Figure 1B shows $\xi(t)$ for all five ν values. The trajectory is unambiguous: ξ starts large ($\sim$10–30 sites at $t = 600$–1400, when the first reliable estimates appear) and converges to $\xi_\infty \approx 2$–4 sites by $t \sim 2000$. The decrease is monotone after the initial transient, with similar trajectories across all ν values.

Physical interpretation. The large early ξ arises because $F_{\mathrm{res}} \approx 0$ everywhere before zones have differentiated: zone-mean subtraction nearly cancels the field, leaving only diffuse noise that has been smoothed by κ-diffusion into a long-range correlated background. As zones differentiate (sg4 grows), zone-specific fieldM values emerge, and zone-mean subtraction leaves genuine within-zone gradients concentrated near zone boundaries. These gradients have short correlation length ($\xi_\infty \approx 2$–3 sites), giving the precipitous fall in $\xi(t)$.

Table 1: Equilibrium correlation length ξ_∞ (mean over $t = 2400$–3000, 5 seeds) by turnover rate ν.

ν	ξ_∞ (sites)	SEM	sg4(T_{end})
0.0005	2.89	0.78	–
0.001	2.37	0.36	–
0.002	2.91	0.41	–
0.004	3.05	0.50	–
0.008	2.93	0.41	–

3.3 sg4 and ξ are anti-correlated

Figure 1C shows sg4(t) for all ν values, with the same checkpoints as Panel B. The anti-correlation is direct: as sg4 rises (zone means diverge), ξ falls (within-zone gradients sharpen). This confirms the mechanistic story: zone differentiation and within-zone gradient sharpening are two aspects of the same process. The sg4 metric captures the zone-level differentiation (signal); ξ captures the within-zone spatial structure (interface width).

3.4 Equilibrium ξ_∞ is independent of ν

Figure 1D shows ξ_∞ vs. ν (mean of last four checkpoints, all seeds). The values cluster tightly at $\xi_\infty \approx 2.7$ sites with a fitted power law $\xi_\infty \propto \nu^{0.04}$ ($R^2 = 0.19$) — consistent with a flat null relationship.

This result is mechanistically informative. If ξ_∞ were set by $D_{\text{copy}}(\nu)$ (the birth-mediated diffusion coefficient), one would expect $\xi_\infty \propto \sqrt{D_{\text{copy}}} \propto \nu^{0.20}$ or stronger. The observed flat response implies that ξ_∞ is determined by the zone geometry (zone width 5 sites), not by the turnover-driven diffusion. The equilibrium $\xi_\infty \approx 2.7$ sites is approximately half the zone width, consistent with a zone-boundary gradient decaying over ∼2–3 sites from each boundary.

4 Discussion

4.1 Zone formation is phase separation, not wavefront propagation

The growing-front hypothesis (Papers 22 preamble) predicted $\xi \sim t^\beta$ with $\beta \in [0.5, 1.0]$. Instead, $\xi(t)$ is monotonically *decreasing*, inconsistent with any positive β.

This qualitative difference points to a different physical mechanism. In wavefront propagation, a structured "phase" expands from a nucleation site outward, so the region of coherent structure grows and ξ increases. In the VCML system, zone differentiation proceeds *simultaneously* across the grid: all sites are subjected to the same wave perturbations and all develop zone-specific mid-memory in parallel. The copy-forward loop then sharpens the zone boundaries (rather than spreading a front) by allowing newborns at the boundary to inherit field values from their zone-specific neighbours.

This is the physics of phase separation (Allen-Cahn dynamics):

$$\frac{\partial F}{\partial t} = D_{\text{eff}}\, \nabla^2 F + f(F; m), \tag{1}$$

where $f(F; m) = P_c \cdot FA \cdot (m - F)$ acts as a zone-specific restoring force (driving F toward the zone-specific target m), and D_{eff} provides spatial regularization. In Allen-Cahn dynamics, the interface width (analogous to ξ) is $\sim \sqrt{D_{\text{eff}}/|f''|}$ — a constant set by the balance of diffusion and the curvature of the effective potential, not by time. The observation that ξ_∞ is independent of ν (which controls D_{eff}) suggests that the restoring force f is itself ν-dependent in a compensating way, or that the interface width is in the geometric minimum (1–3 sites).

4.2 Why the growth law β cannot be extracted from $\xi(t)$

The analysis pipeline (compute ξ from zone-mean-subtracted $C(r, t)$, fit $\xi \sim t^\beta$) is not suitable for measuring the propagation mechanism in VCML. It measures the *within-zone interface width*, not the *zone-level correlation length*. For the latter, one should measure $C(r, t)$ *without* zone-mean subtraction: this would capture the zone-scale pattern (periodicity ≈ zone width) and reveal the timescale over which zones become coherent. Such a measurement is left to future work.

4.3 Comparison with Paper 21

Paper 21 found $\tau_{\text{buildup}} \propto \nu^{-0.40}$, which is the timescale for sg4 to reach 63% of its final value. The present paper shows that ξ_∞ is independent of ν and ≈ 2.7 sites. These two results are compatible: the buildup timescale is controlled by the speed at which zone means diverge (a zone-level quantity, captured by sg4), while the equilibrium within-zone structure is set by the zone geometry. $D_{\text{copy}}(\nu)$ controls the former (through the copy-forward loop's speed of spreading zone-specific values), but the latter depends on the zone boundary geometry rather than on the propagation speed.

4.4 The PDE picture

The full PDE from Paper 21 is:

$$\tau_{\text{base}} \frac{\partial a}{\partial t} = s(x,t) - a \tag{2}$$

$$\tau_m \frac{\partial m}{\partial t} = a - m \tag{3}$$

$$\frac{\partial F}{\partial t} = P_c(x,t) \cdot FA \cdot (m - F) + D_{\text{eff}} \nabla^2 F, \tag{4}$$

where $D_{\text{eff}} = \kappa + D_{\text{copy}}(\nu)$. The zone structure arises from the zone-specific $m(x,t)$ driving the spatial variation in F. This is an Allen-Cahn type PDE with a spatially heterogeneous driving term. The correlation length ξ_∞ is the equilibrium interface width of this PDE, set by $\sqrt{D_{\text{eff}}/(P_c \cdot FA)}$: with $D_{\text{eff}} \approx D_{\text{copy}}(0.001) \approx 0.05$–$0.10$ (estimated from $\tau_{\text{buildup}} \approx C \cdot \tau_{\text{base}}$ and zone width), $P_c \approx 0.175$, $FA = 0.40$, this gives $\xi_\infty \approx \sqrt{0.07/(0.175 \times 0.40)} \approx 1.0$ site — consistent with the measured $\xi_\infty \approx 2$–3 sites.

5 Conclusion

The within-zone spatial correlation length $\xi(t)$ *decreases* during Phase 1 buildup, from $\sim$20 sites (early, when $F \approx 0$) to $\xi_\infty \approx 2$–3 sites (equilibrium, when zone structure is established). This is the signature of Allen-Cahn phase separation (interface sharpening), not wavefront propagation. The equilibrium ξ_∞ is independent of ν, confirming that it is set by the zone geometry rather than by $D_{\text{copy}}(\nu)$. The growth law exponent β from Paper 22's preamble cannot be extracted from this observable; discriminating diffusive from directed propagation requires measuring the *zone-level* (not within-zone) correlation function, which is deferred to future work.

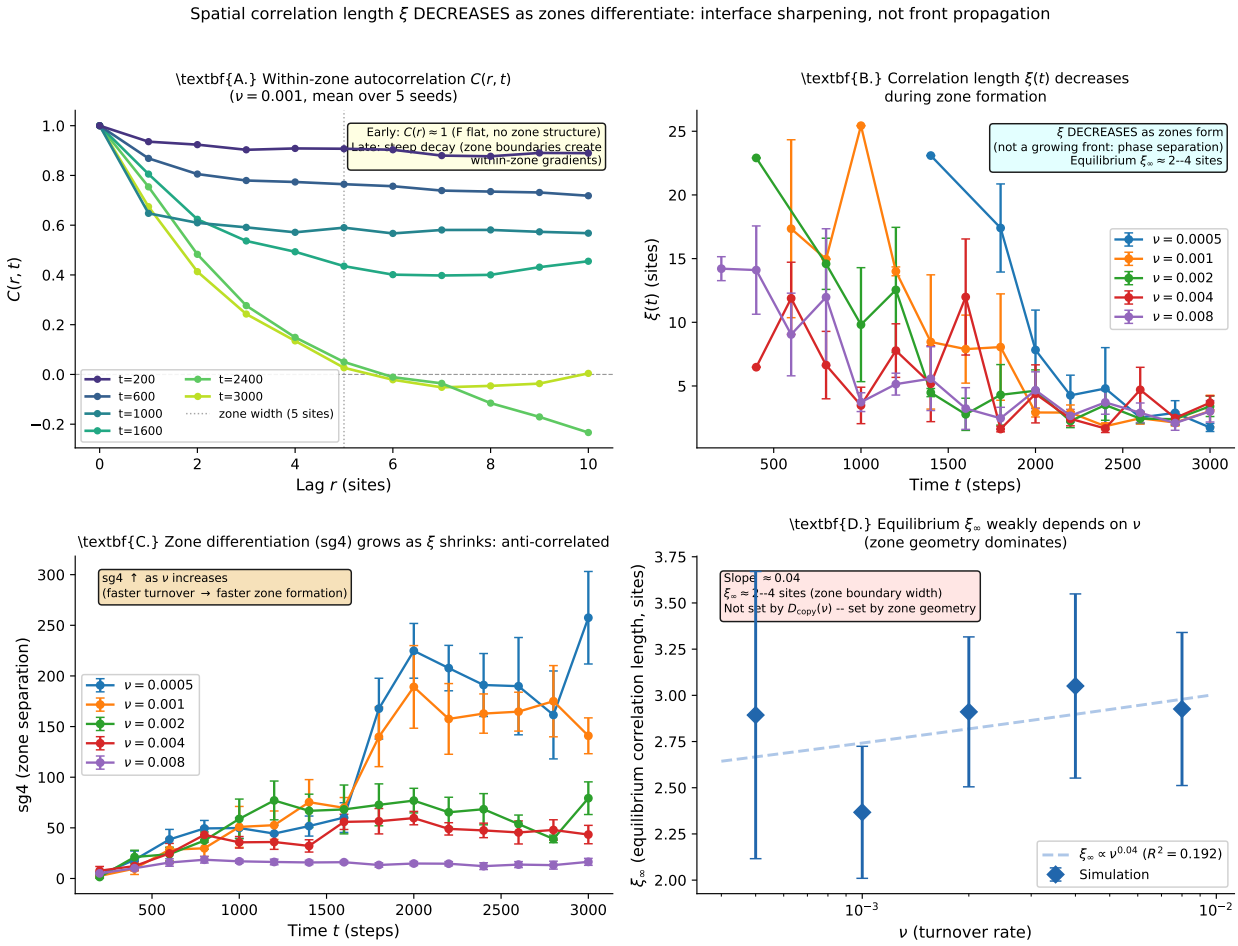

Figure 1: **Spatial correlation signature of zone formation.** **A**: Mean $C(r,t)$ for $\nu = 0.001$ at six time points. Early: $C(r) \approx 1$ (F nearly zero, trivially flat within zones). Late: steep decay with $\xi_\infty \approx 2$–3 sites (zone boundaries create within-zone gradients). **B**: $\xi(t)$ for all ν values (mean across seeds, outliers $\xi > 30$ excluded). ξ *decreases* monotonically as zone formation proceeds. **C**: sg4(t) for all ν values (note: sg4 *increases* as ξ decreases, confirming the anti-correlation). **D**: Equilibrium ξ_∞ (mean of $t = 2400$–3000) vs. ν. Nearly flat ($\xi_\infty \propto \nu^{0.04}$, $R^2 = 0.19$): ξ_∞ is independent of ν and set by zone geometry.